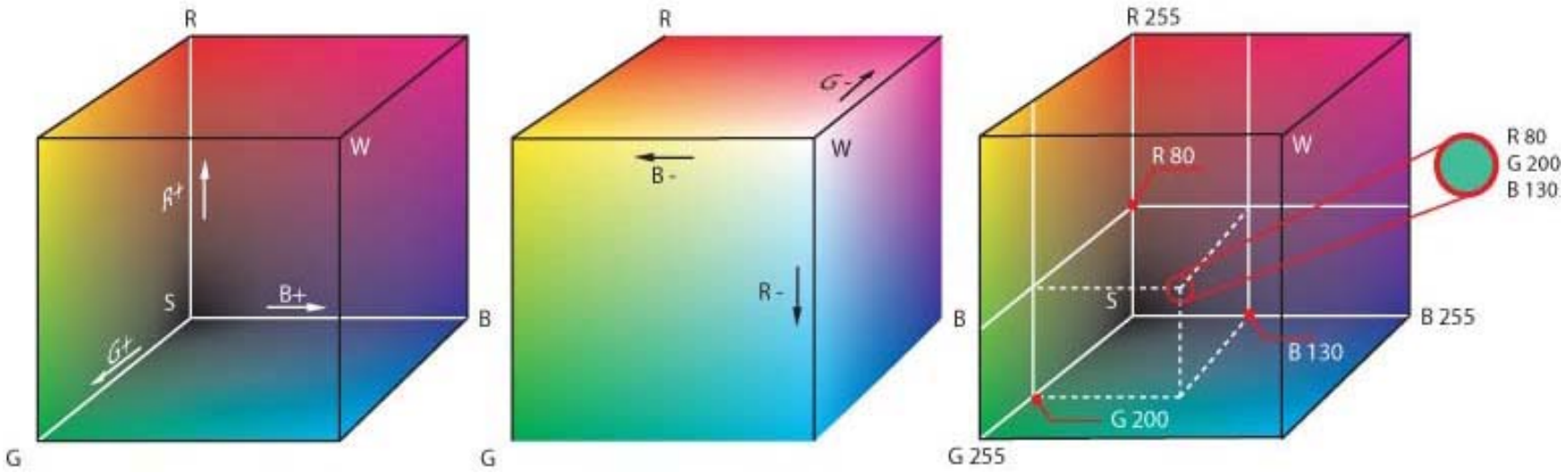
14. Color Mapping

Recall: RGB color model

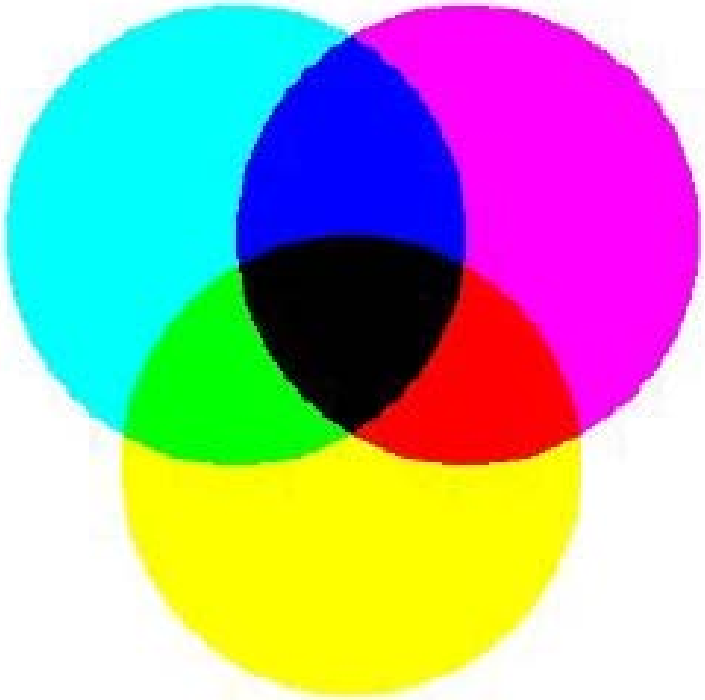


CMY color model

- The CMY color model is related to the RGB color model.
- Its base colors are
  - cyan (C)
  - magenta (M)
  - yellow (Y)
- They are arranged in a 3D Cartesian coordinate system.
- The scheme is subtractive.



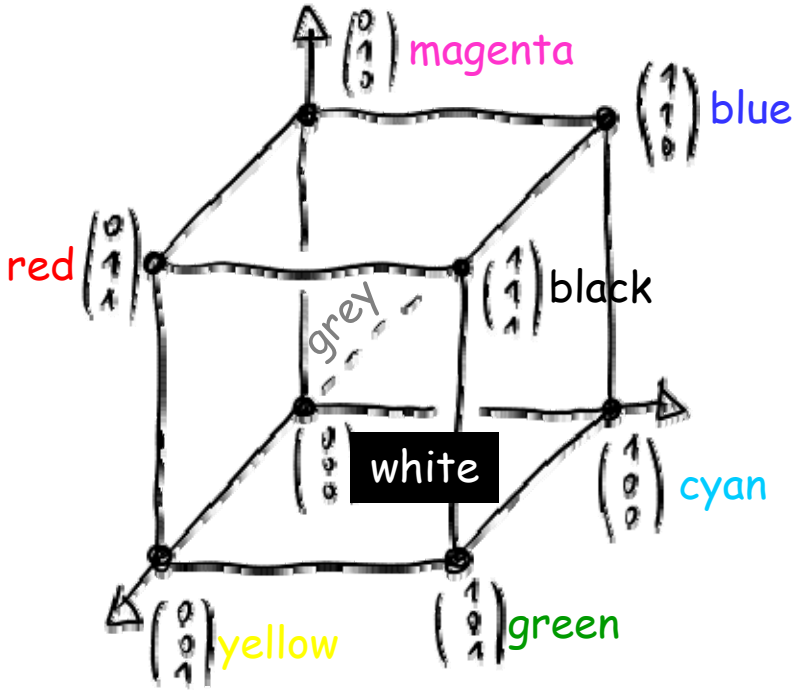
Subtractive color scheme



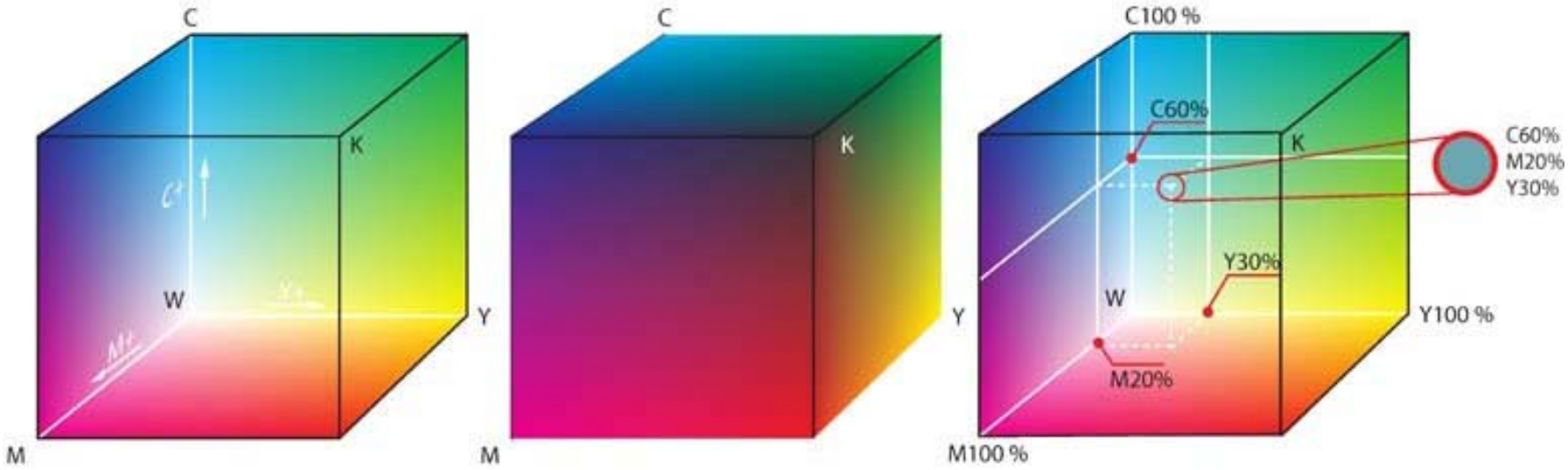
- CMY color model is subtractive, i.e., adding colors makes the resulting color darker.
- Application: color printers.
- As it only works perfectly in theory, typically a black cartridge is added in practice (CMYK color model).

CMY color cube

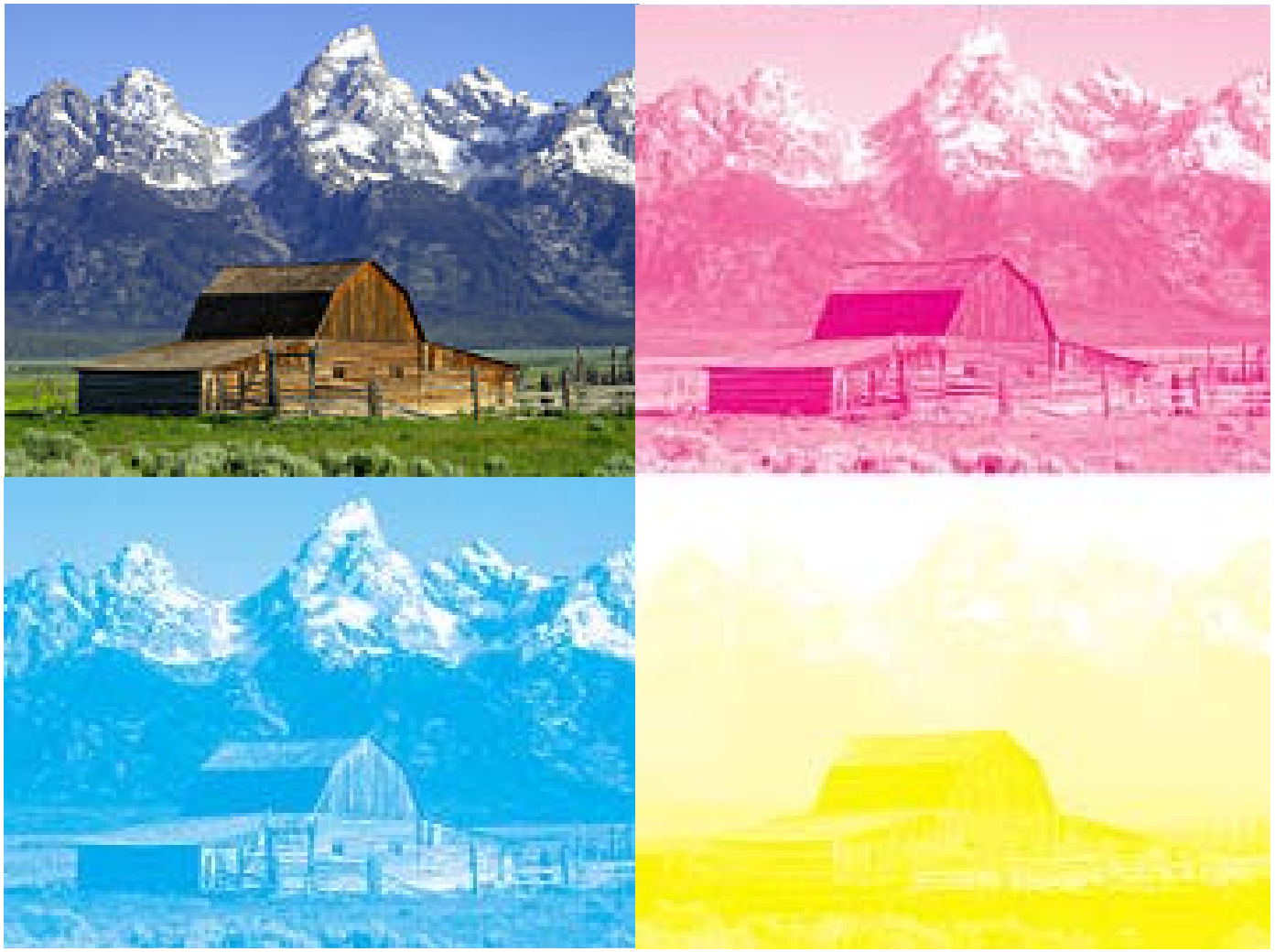
- All colors c that can be generated are represented by the unit cube in the 3D Cartesian coordinate system.



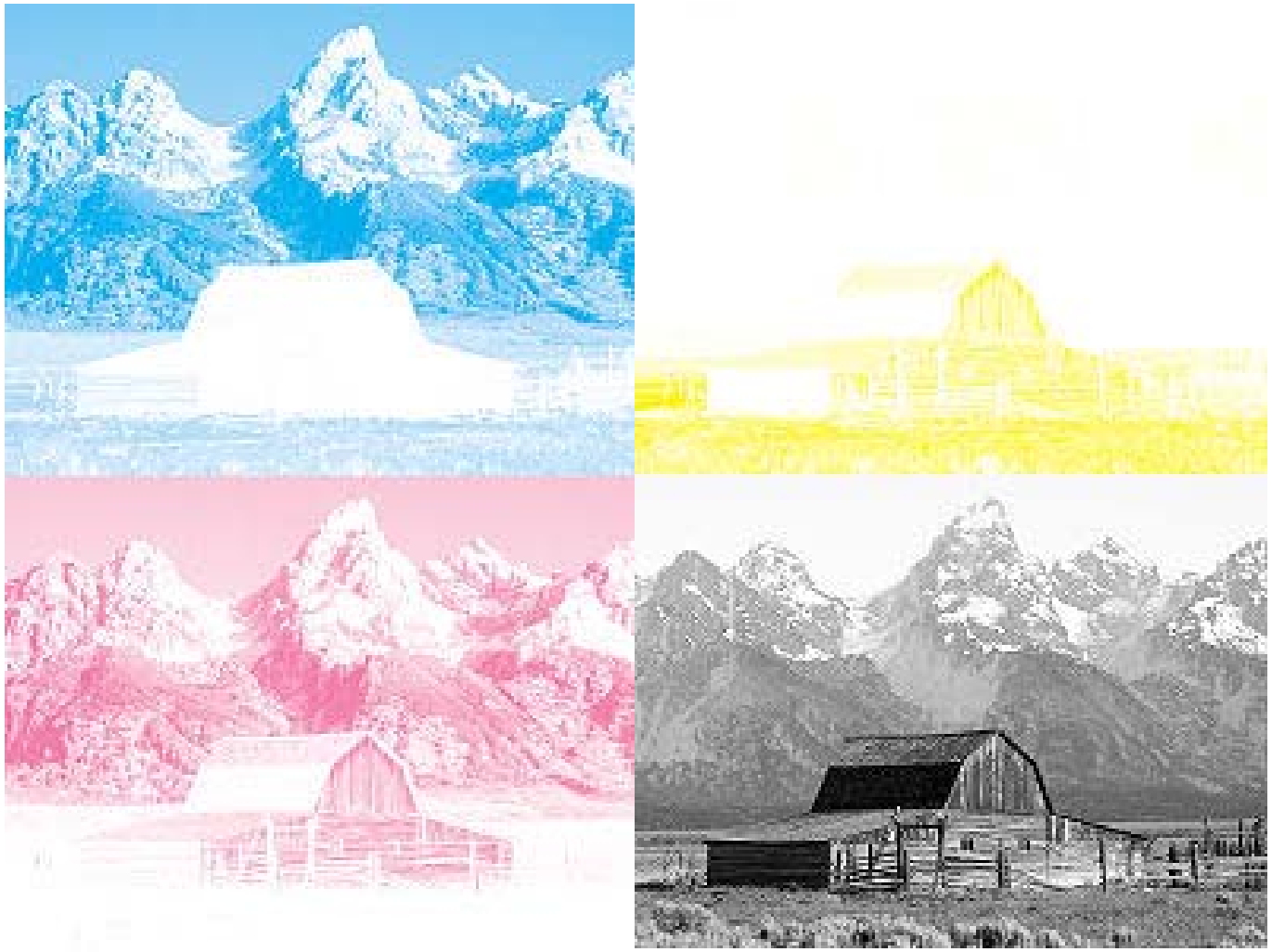
CMY color cube



CMY color model



CMYK color model



Conversion

- RGB -> CMY:

$$\begin{pmatrix} C \\ M \\ Y \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - \begin{pmatrix} R \\ G \\ B \end{pmatrix}$$

- CMY -> RGB:

$$\begin{pmatrix} R \\ G \\ B \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - \begin{pmatrix} C \\ M \\ Y \end{pmatrix}$$



Conversion

- CMY -> CMYK:

$$\begin{pmatrix} C \\ M \\ Y \\ K \end{pmatrix} = \begin{pmatrix} C - \min\{C, M, Y\} \\ M - \min\{C, M, Y\} \\ Y - \min\{C, M, Y\} \\ \min\{C, M, Y\} \end{pmatrix}$$

- CMYK -> CMY:

$$\begin{pmatrix} C \\ M \\ Y \end{pmatrix} = \begin{pmatrix} C + K \\ M + K \\ Y + K \end{pmatrix}$$



HSV color model

- While RGB and CMY color models have their application in hardware implementations, the HSV color model is based on properties of human perception.
- Its application is for human interfaces.



HSV color model

The HSV color model also consists of 3 channels:

- H: When perceiving a color, we perceive the dominant wavelength. This is represented by the **hue** (H).
- S: The purity of a color is measured by the amount of frequencies in the light. The smaller the frequency spectrum, the purer the color. This is represented by the **saturation** (S).
- V: The maximum amplitude of the light is given at its dominant wavelength. It represents the energy of the light given in form of its **value** (V).



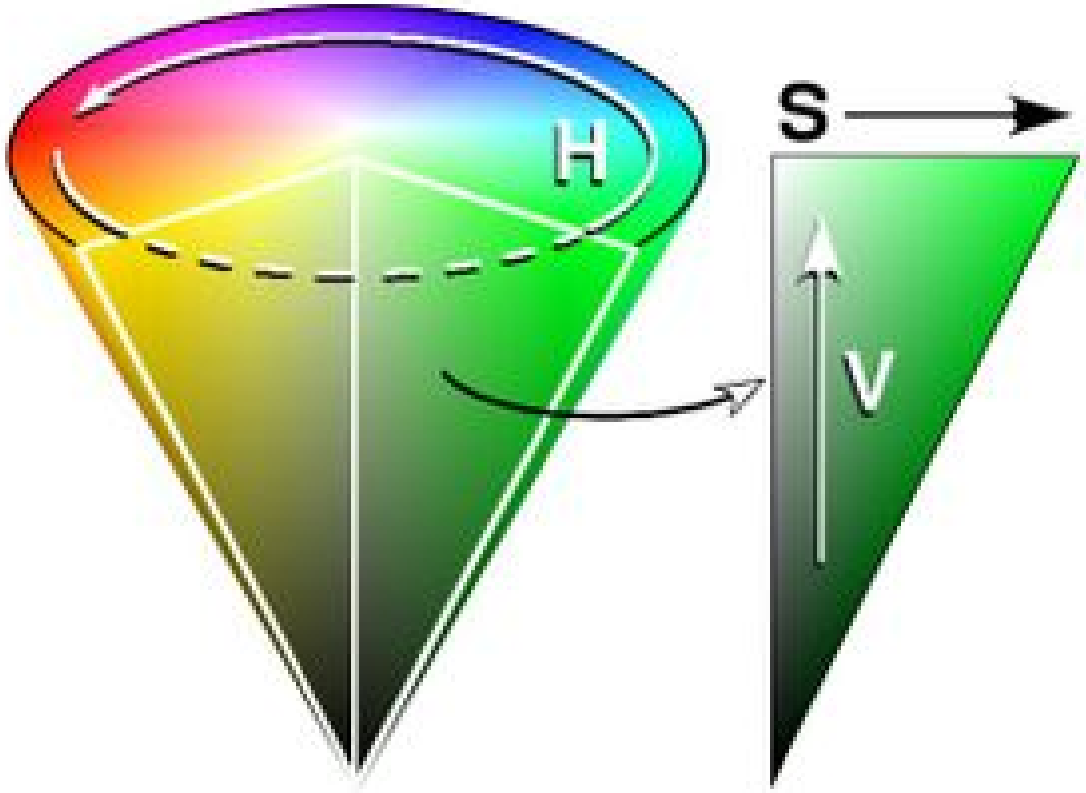
HSV coordinate system

- The coordinate system of the HSV color model is given in form of a cone:
  - H is given in form of an angle in the range [0,360) which represents the rotational symmetry of the cone.

    The order of the colors is given by the frequency spectrum.
  - S is in the range [0,1] and its axis perpendicular to the V axis.
  - V is in the range [0,1] and its axis is the rotational symmetry axis of the cone.

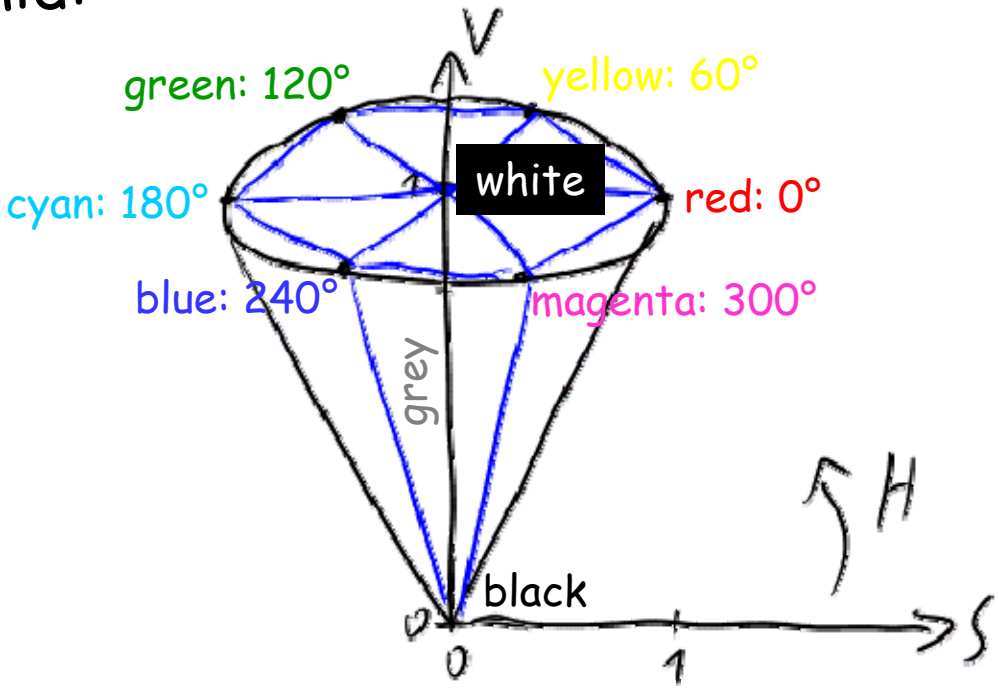


HSV coordinate system



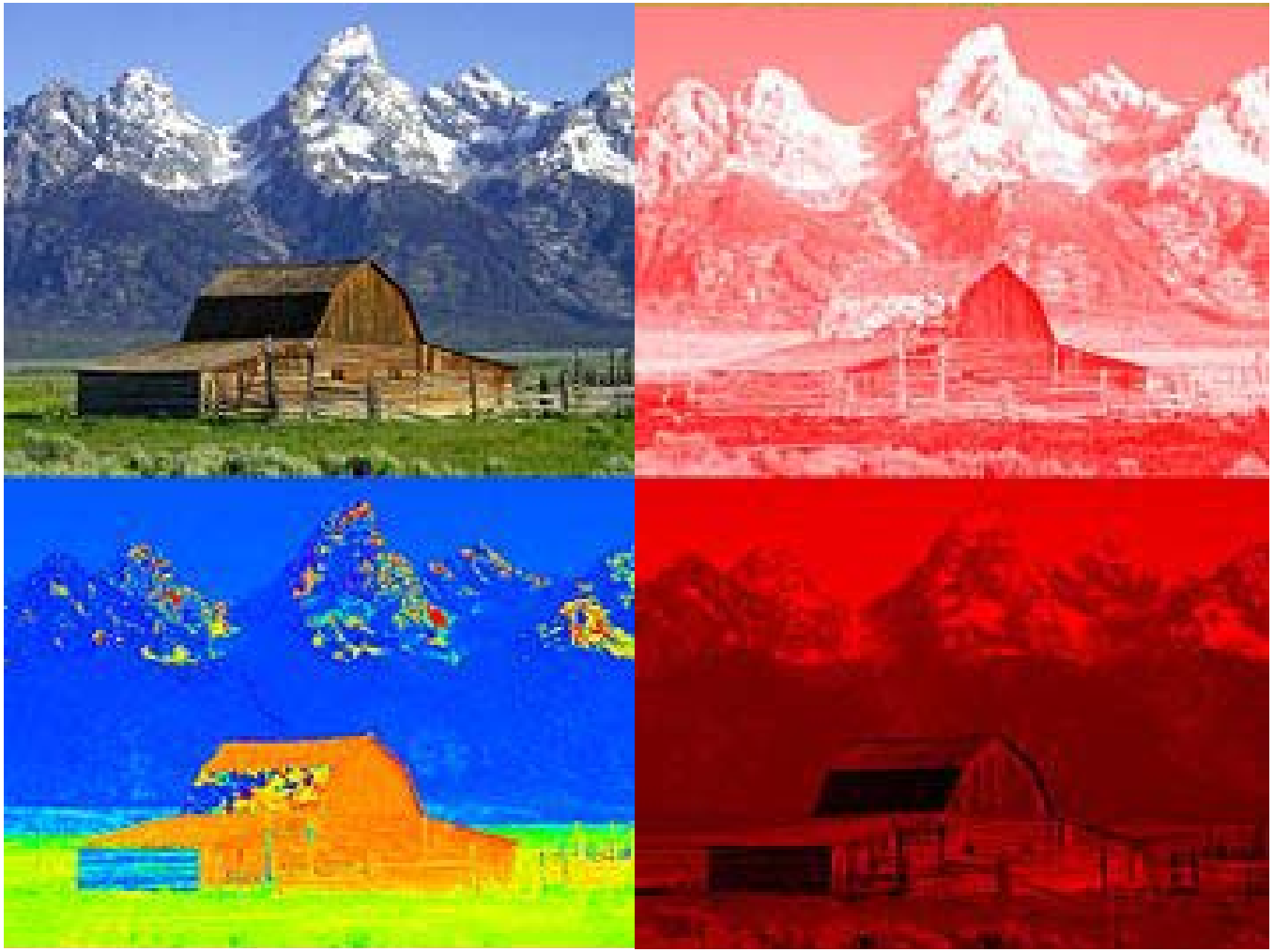
HSV coordinate system

- Sometimes the cone is approximated by a 6-sided pyramid:



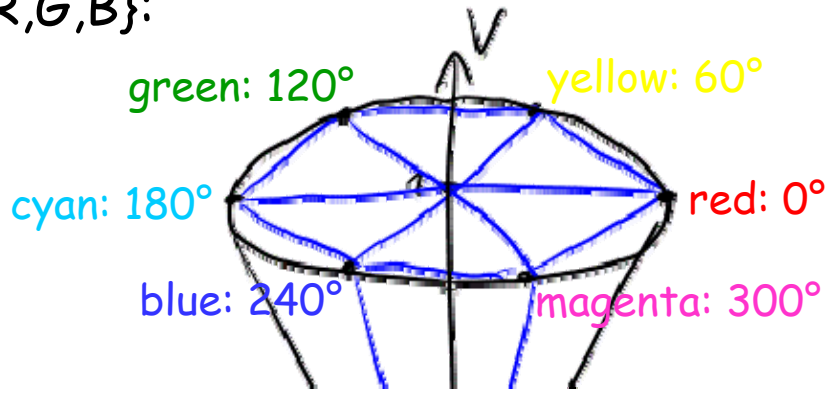
- The hexagon at the base of the 6-sided pyramid is the RGB cube projected along the grey axis.

HSV color model



Conversion: RGB -> HSV

Let max = max {R,G,B} and min = min {R,G,B}:



$$v = \max$$

$$s = \begin{cases} 0, & \text{if } \max = 0 \\ \frac{\max - \min}{\max} = 1 - \frac{\min}{\max}, & \text{otherwise} \end{cases}$$

$$h = \begin{cases} 0 & \text{if } \max = \min \\ (60° \times \frac{g-b}{\max - \min} + 0°) \bmod 360°, & \text{if } \max = r \\ 60° \times \frac{b-r}{\max - \min} + 120°, & \text{if } \max = g \\ 60° \times \frac{r-g}{\max - \min} + 240°, & \text{if } \max = b \end{cases}$$

Conversion HSV -> RGB

$$h_i = \left\lfloor \frac{h}{60} \right\rfloor \mod 6$$

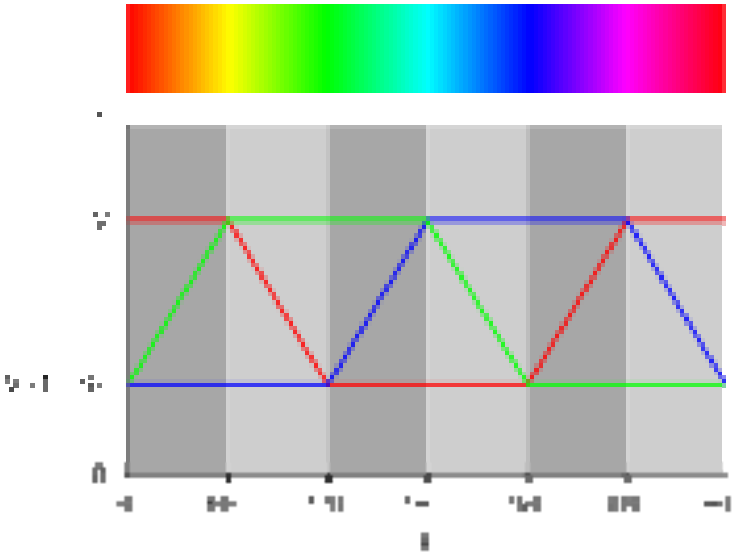
$$f = \frac{h}{60} - \left\lfloor \frac{h}{60} \right\rfloor$$

$$p = v \times (1 - s)$$

$$q = v \times (1 - f \times s)$$

$$t = v \times (1 - (1 - f) \times s)$$

$$(r, g, b) = \begin{cases} (v, t, p), & \text{if } h_i = 0 \\ (q, v, p), & \text{if } h_i = 1 \\ (p, v, t), & \text{if } h_i = 2 \\ (p, q, v), & \text{if } h_i = 3 \\ (t, p, v), & \text{if } h_i = 4 \\ (v, p, q), & \text{if } h_i = 5 \end{cases}$$



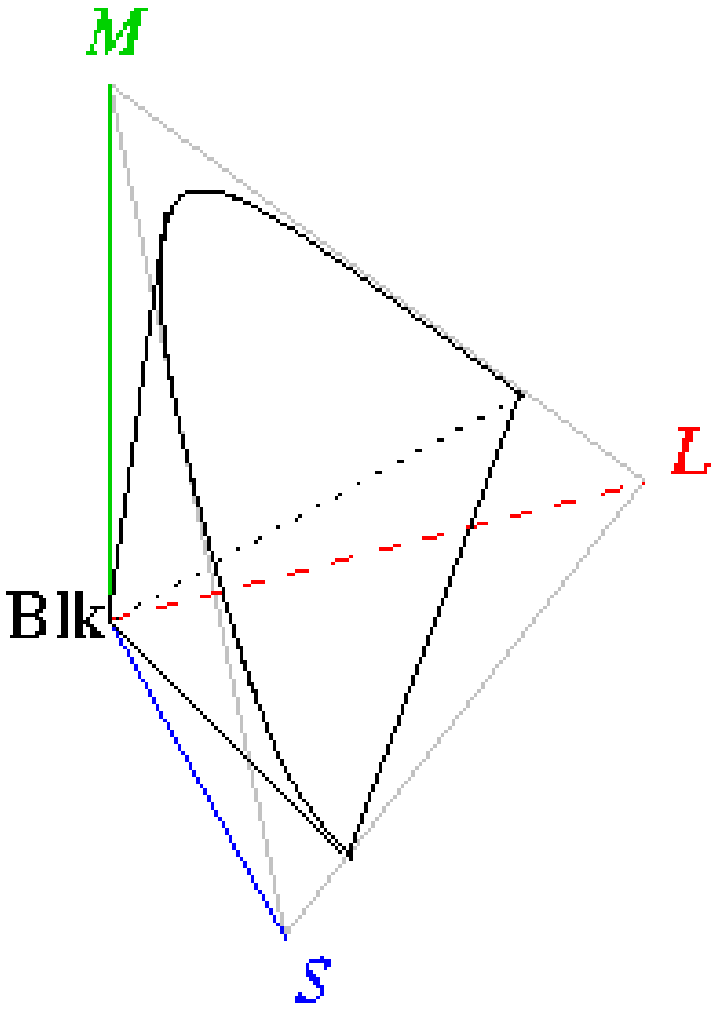
CIE color models

- CIE: Commission Internationale d' Eclairage

- Human color space (tristimulus):
  - *x*, *y*, and *z* axes represent the stimuli for the long-wavelength (*L*), medium-wavelength (*M*), and short-wavelength (*S*) receptors.
  - The human color space is a horse-shoe-shaped cone.
  - The origin corresponds to black and is the tip of the cone.
  - Brighter colors are farther from the origin.
  - The most saturated colors are located at the outer rim of the cone.



Human color space

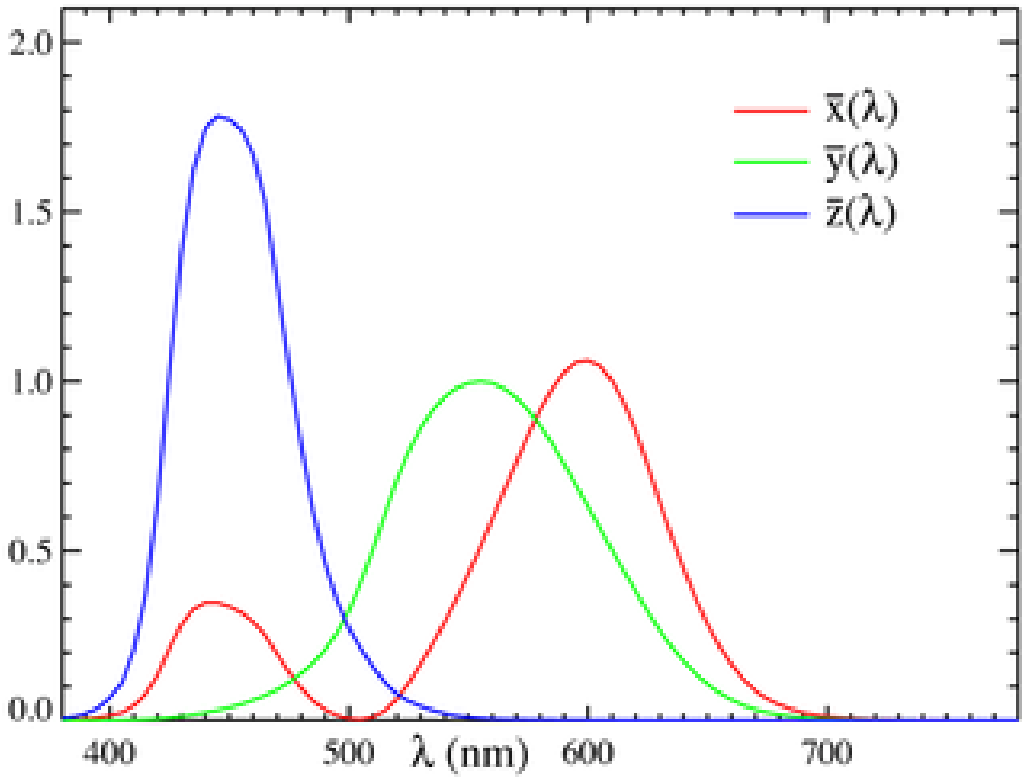


Gamut

- The set of colors described by a color model, is called its gamut.
- We observed that the RGB gamut is a real subset of the human tristimulus gamut.
- CIE tried to overcome this problem by replacing the R, G, and B wavelengths with the tristimulus wavelengths.
- They define chromaticity curves with no negative components.
- The 3 color channels are called X, Y, and Z.
- The set-up is such that Y represents luminance.

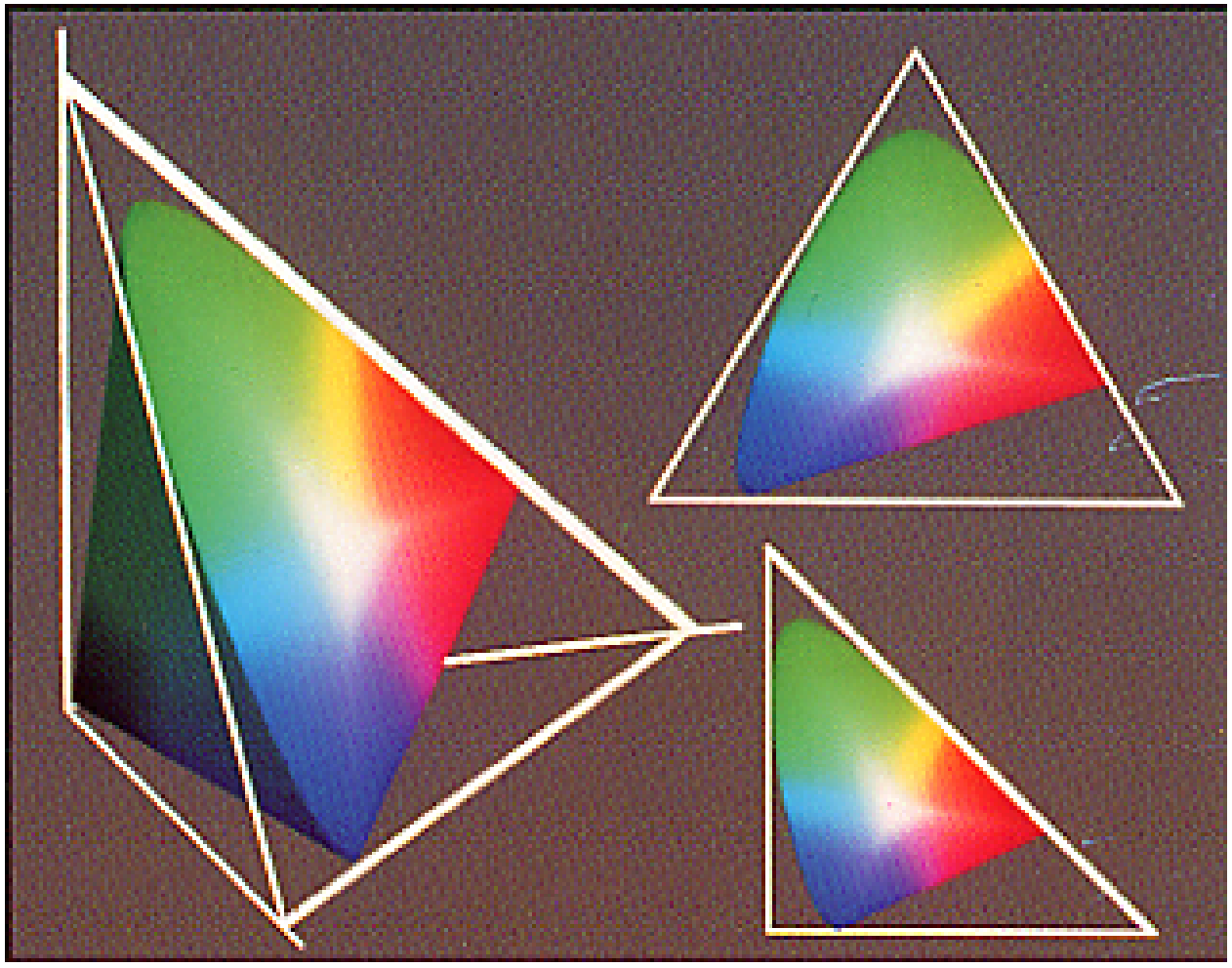


CIE XYZ color model



- The amplitudes of the curve were rather arbitrary.
- The CIE xyY color model modifies them by scaling to equal areas under the curves.

CIE XYZ Color Model



Conversion

- RGB -> CIE XYZ:

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \frac{1}{0.17697} \begin{bmatrix} 0.49 & 0.31 & 0.20 \\ 0.17697 & 0.81240 & 0.01063 \\ 0.00 & 0.01 & 0.99 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

- CIE XYZ -> RGB:
  - inverse matrix
  - caveat: not always all entries will be nonnegative!



CIE xyY color model

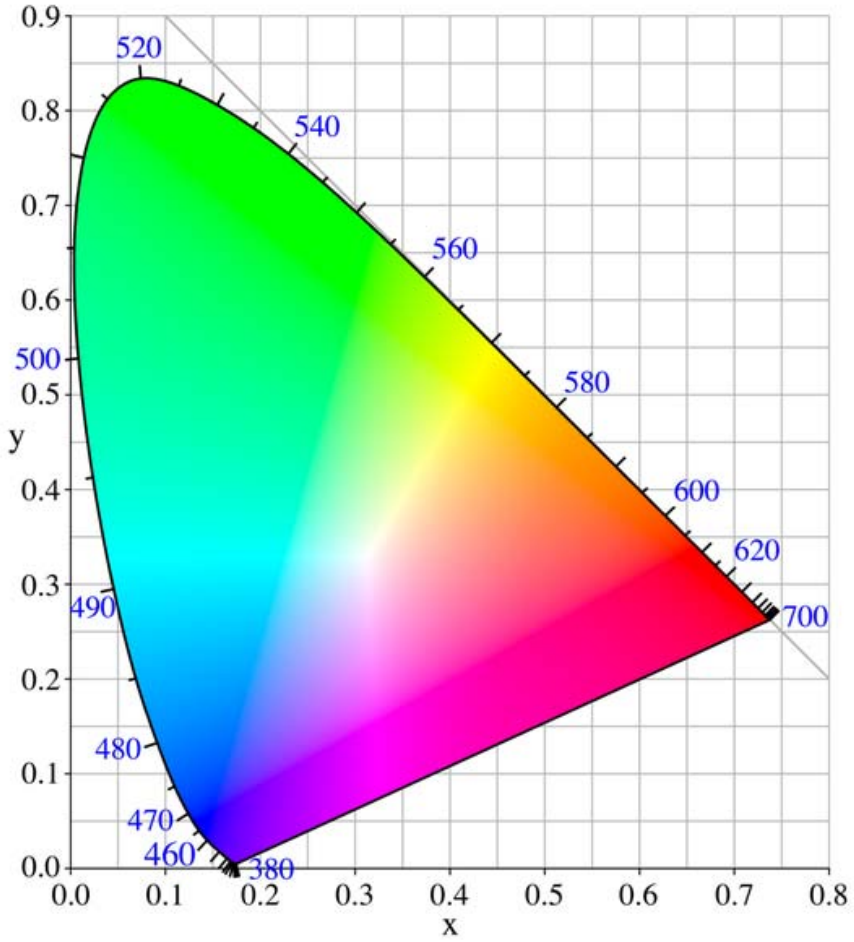
- Normalization:

$$x = \frac{X}{X+Y+Z}$$
$$y = \frac{Y}{X+Y+Z}$$
$$z = \frac{Z}{X+Y+Z} = 1 - x - y$$

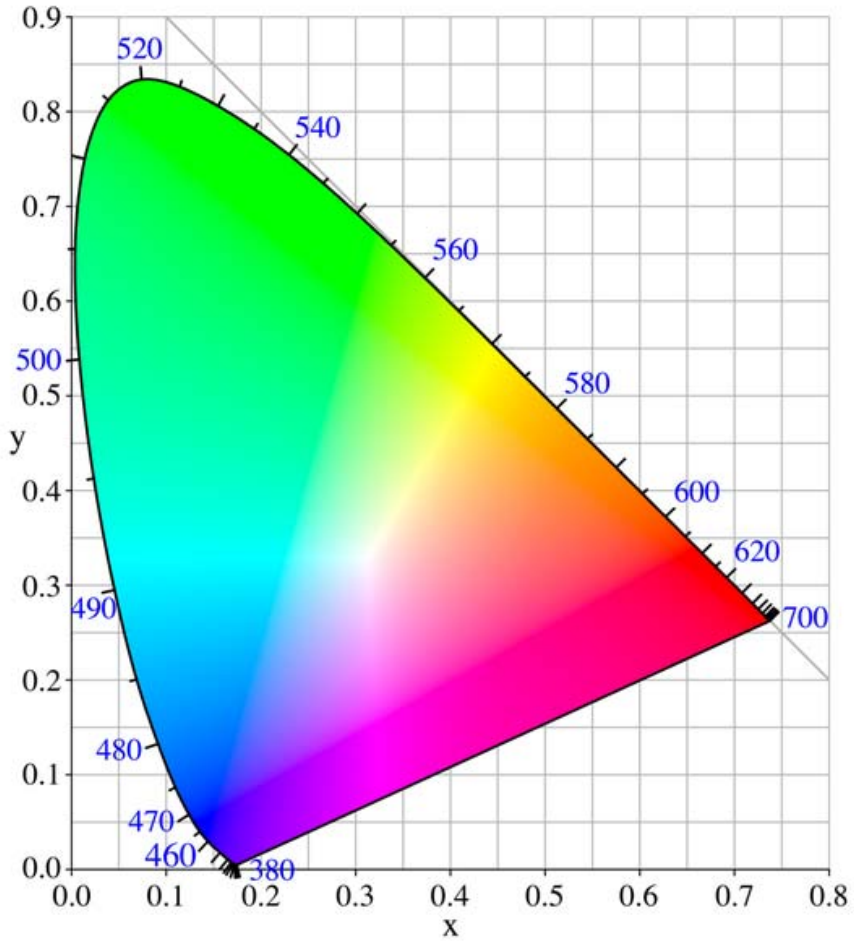
- z is not stored explicitly.
- Instead, we store the luminance channel Y.
- x and y represent the chromatic channels.

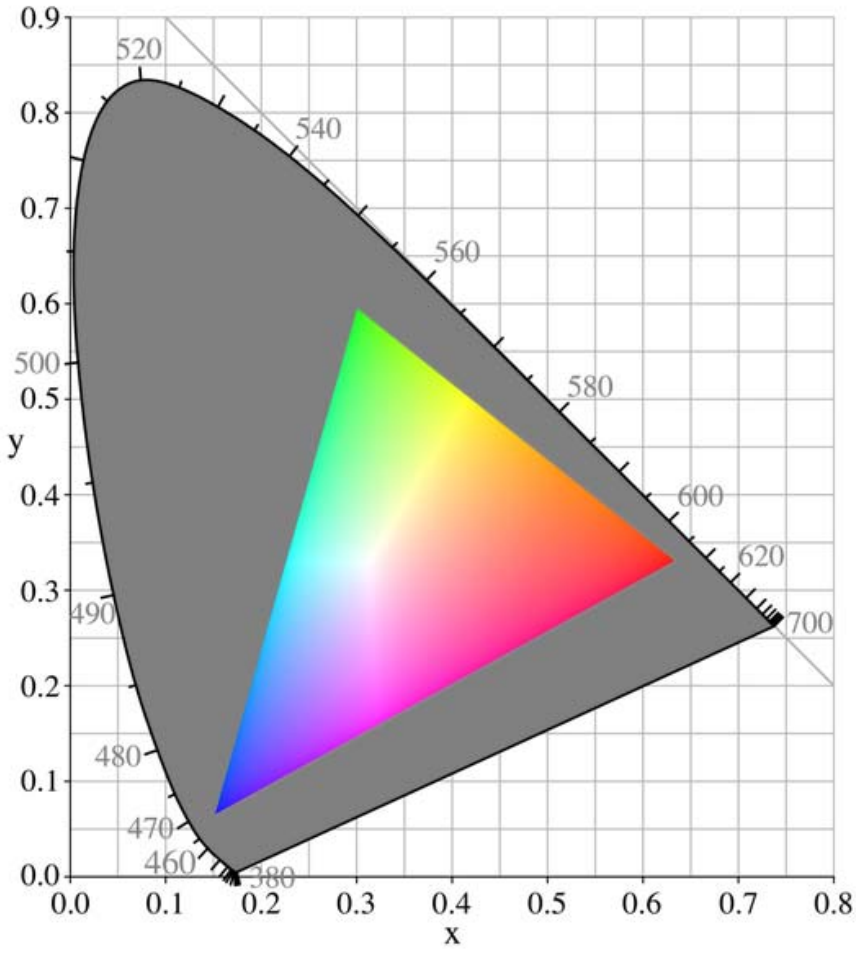


CIE xyY chromaticity diagram



Comparison CIE xyY vs. RGB





CIE L*a*b* color model

- To further strengthen the correlation between the color model and human perception, we apply the following non-linear transformation:

$$L^* = 116\, f(Y/Y_n) - 16$$
$$a^* = 500\,[f(X/X_n) - f(Y/Y_n)]$$
$$b^* = 200\,[f(Y/Y_n) - f(Z/Z_n)]$$

with

$$f(t) = \begin{cases} t^{1/3} & t > (6/29)^3 \\ \frac{1}{3}\left(\frac{29}{6}\right)^2 t + \frac{4}{29} & \text{otherwise} \end{cases}$$

where (X_n, Y_n, Z_n) are the tristimulus values of the reference white point.



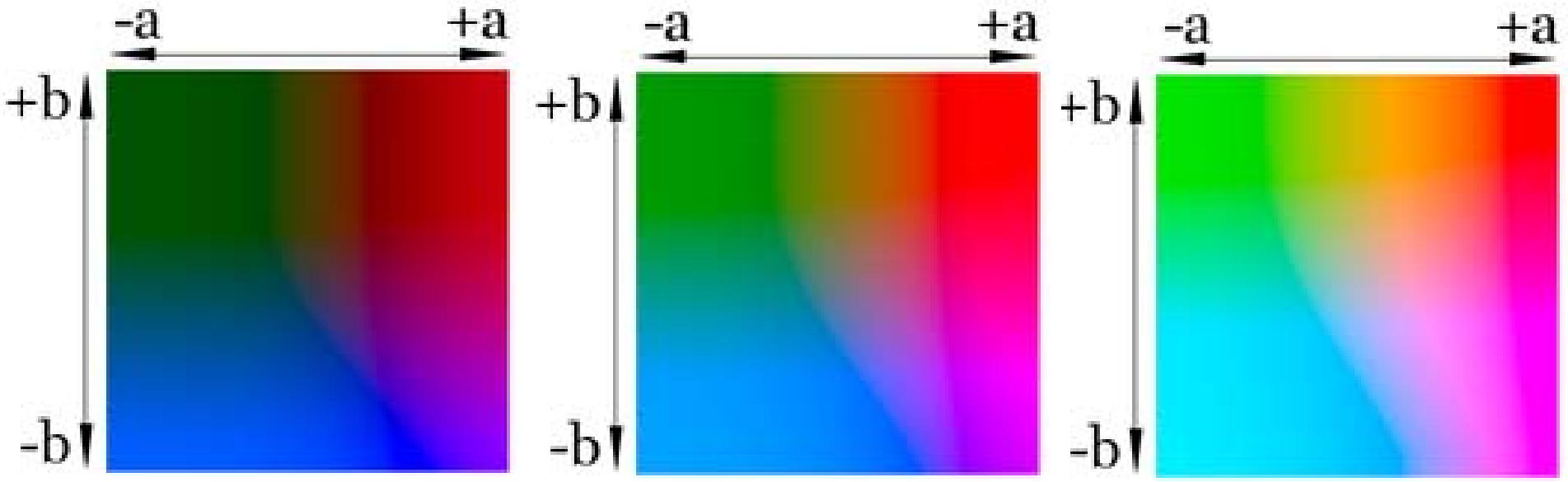
CIE L*a*b* color model

- L* represents a luminance channel (relative to reference white point).
- a* and b* represent chromatic channels.



CIE L*a*b* color model

Chromaticity diagram for various luminances:



CIE L*a*b* color model

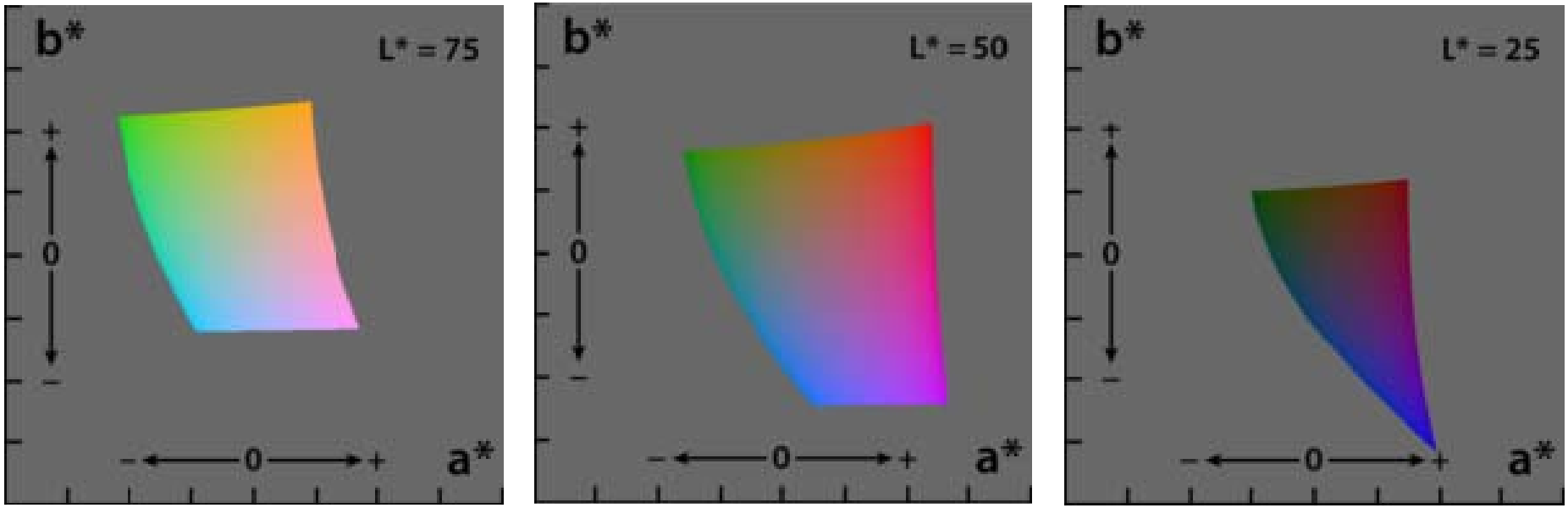
Perceptually uniform:

- The CIE L*a*b* color model is *perceptually uniform*, i.e., a change of the same amount in a color value should produce a change of about the same visual importance.
- In other words, Euclidean distance in the color space is propotional to human perception.



L*a*b* & RGB

- L*a*b* chromaticity diagram reduced to the colors that can be represented in RGB color space.



Summary

- We have introduced additive (RGB) and subtractive (CMY) color schemes, where the channels represent colors as in a tristimulus.
- HSV color scheme is based on channels reflecting visual perception properties.
- CIE XYZ/xyY follow the human tristimulus.
- CIE L*a*b* is perceptionally uniform.



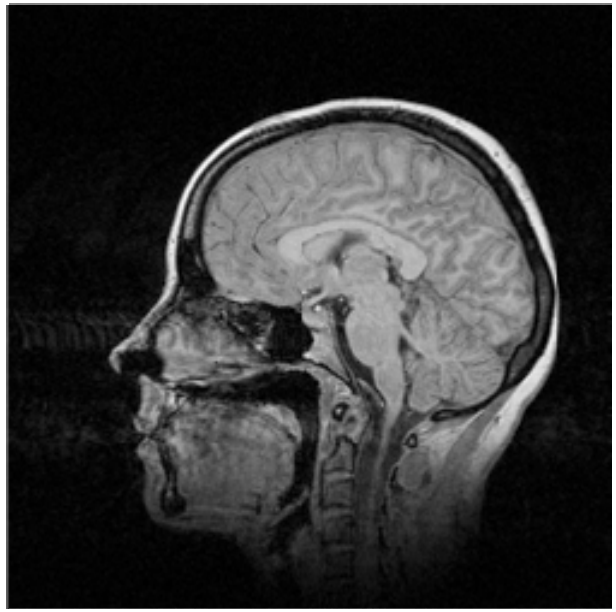
Color map construction

- Example 1: Luminance map

  By assigning to each of the RGB color channels the same values one exclusively obtains greyscale colors.

  A linear mapping of numercial values to greyscale values generates a luminance map.

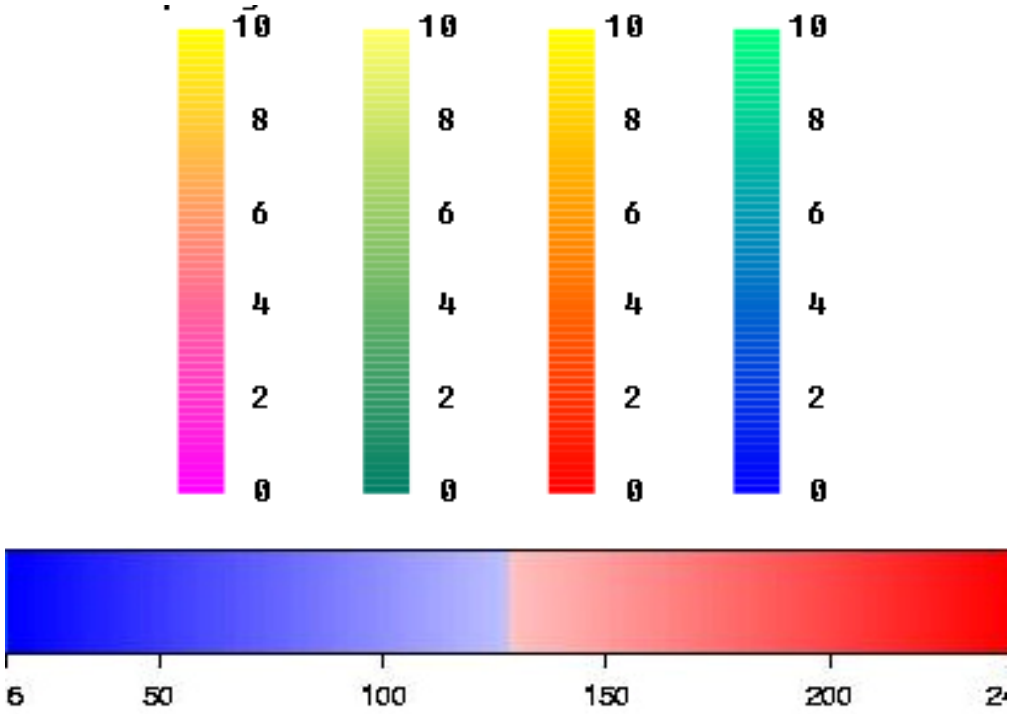




Color map construction

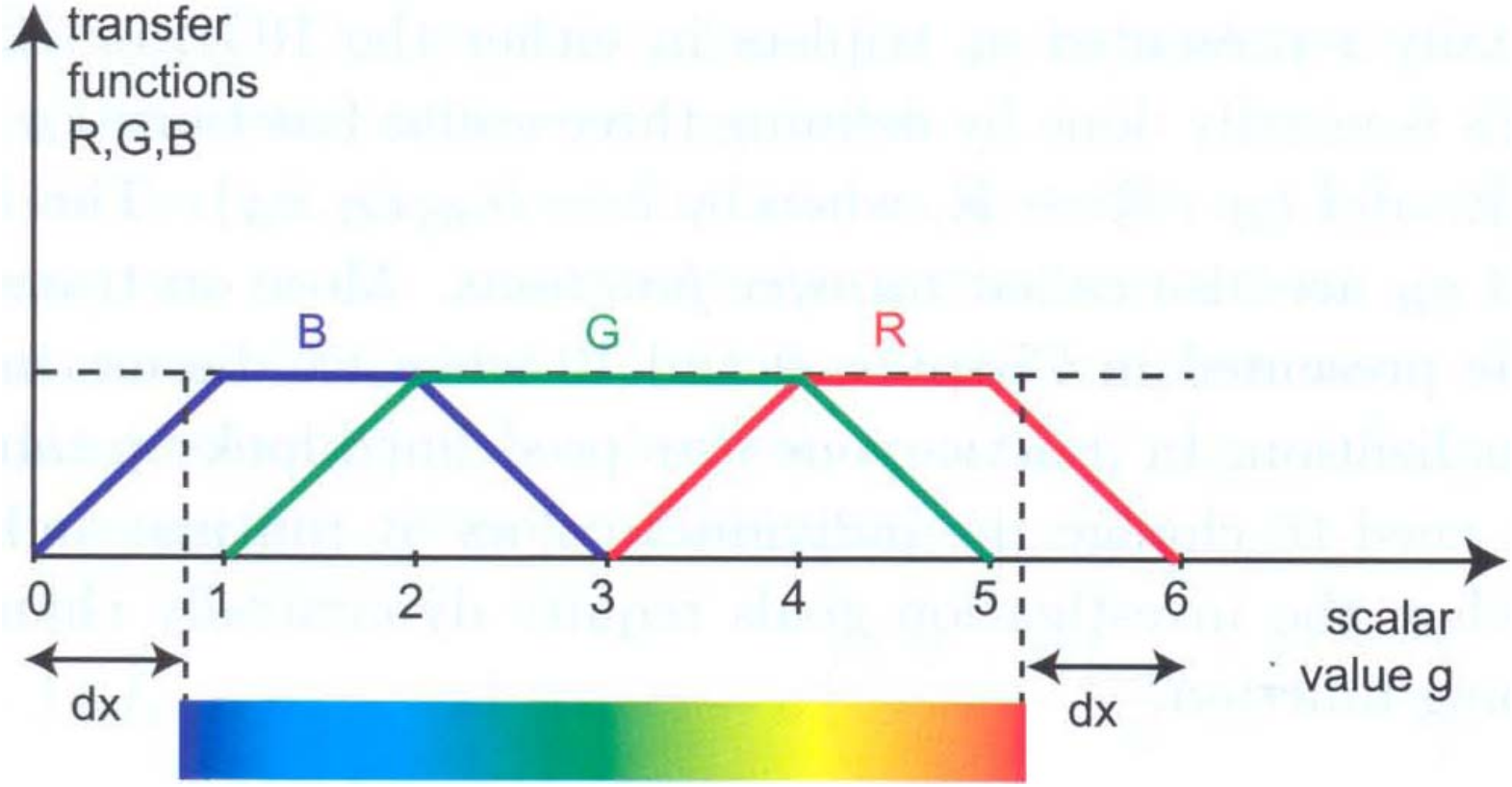
- Example 2: Linear color transition

  One can steadily decrease one color's contribution and increase another's to produce color transitions.



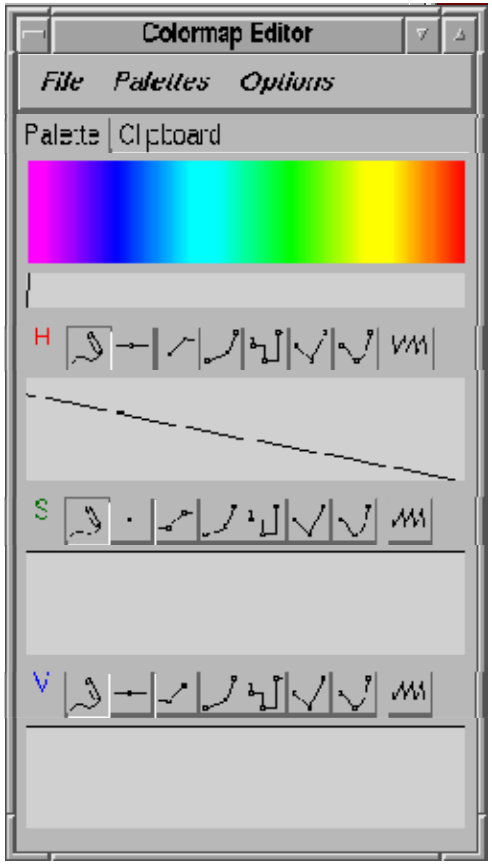
Color map construction

- Example 3: Rainbow color map

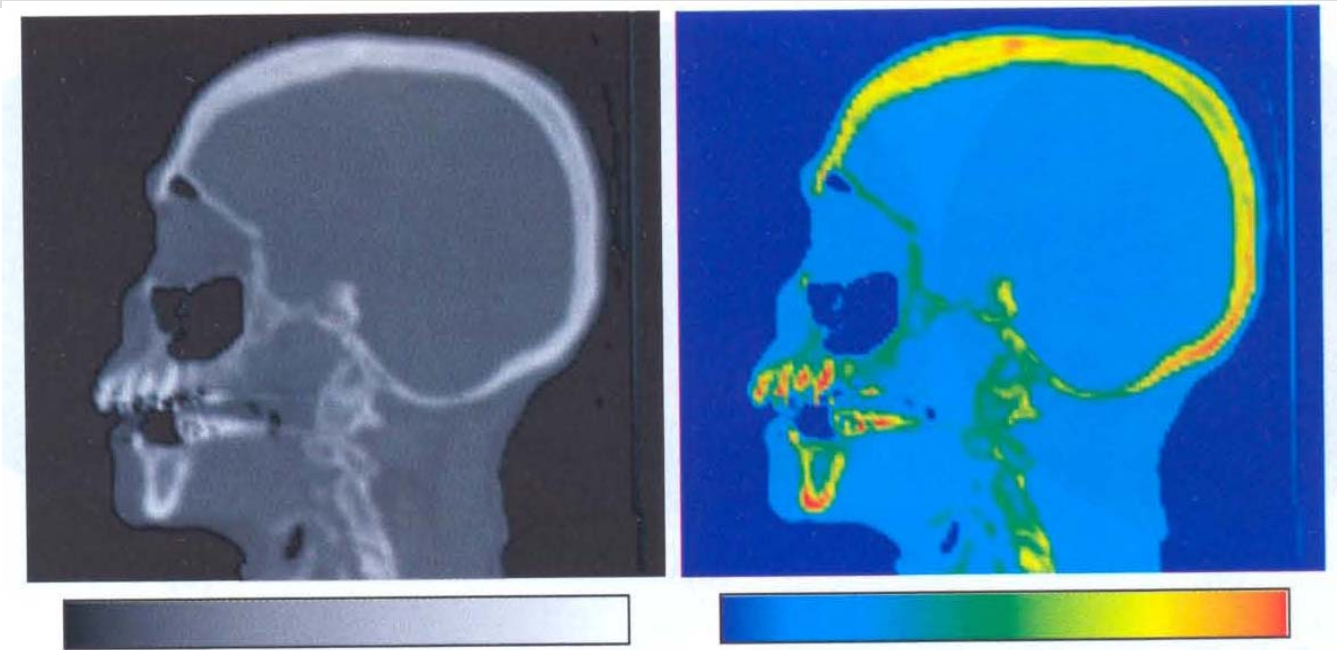


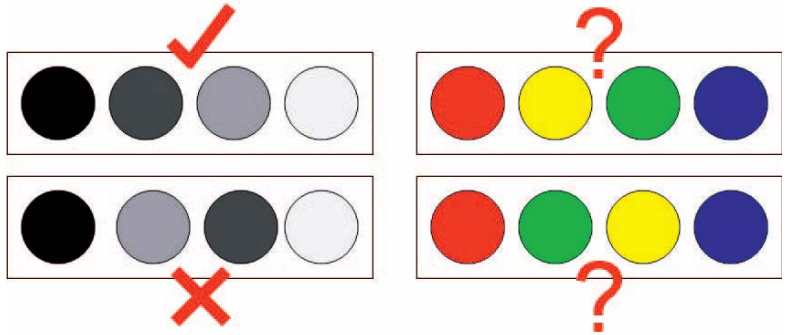
Color map construction

- More intuitively, the rainbow color map can be generated using the HSV color space:



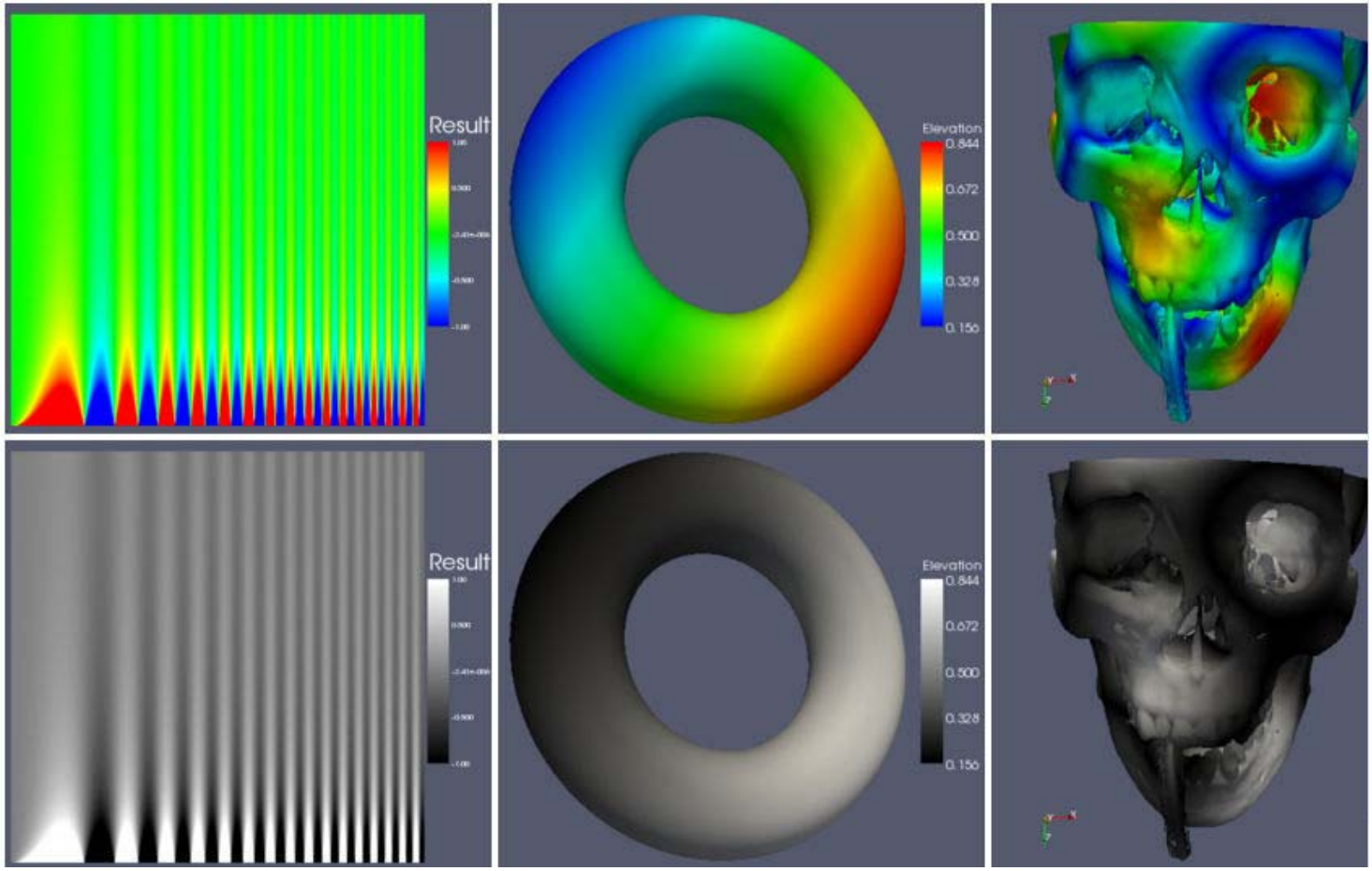
Luminance vs. rainbow color map





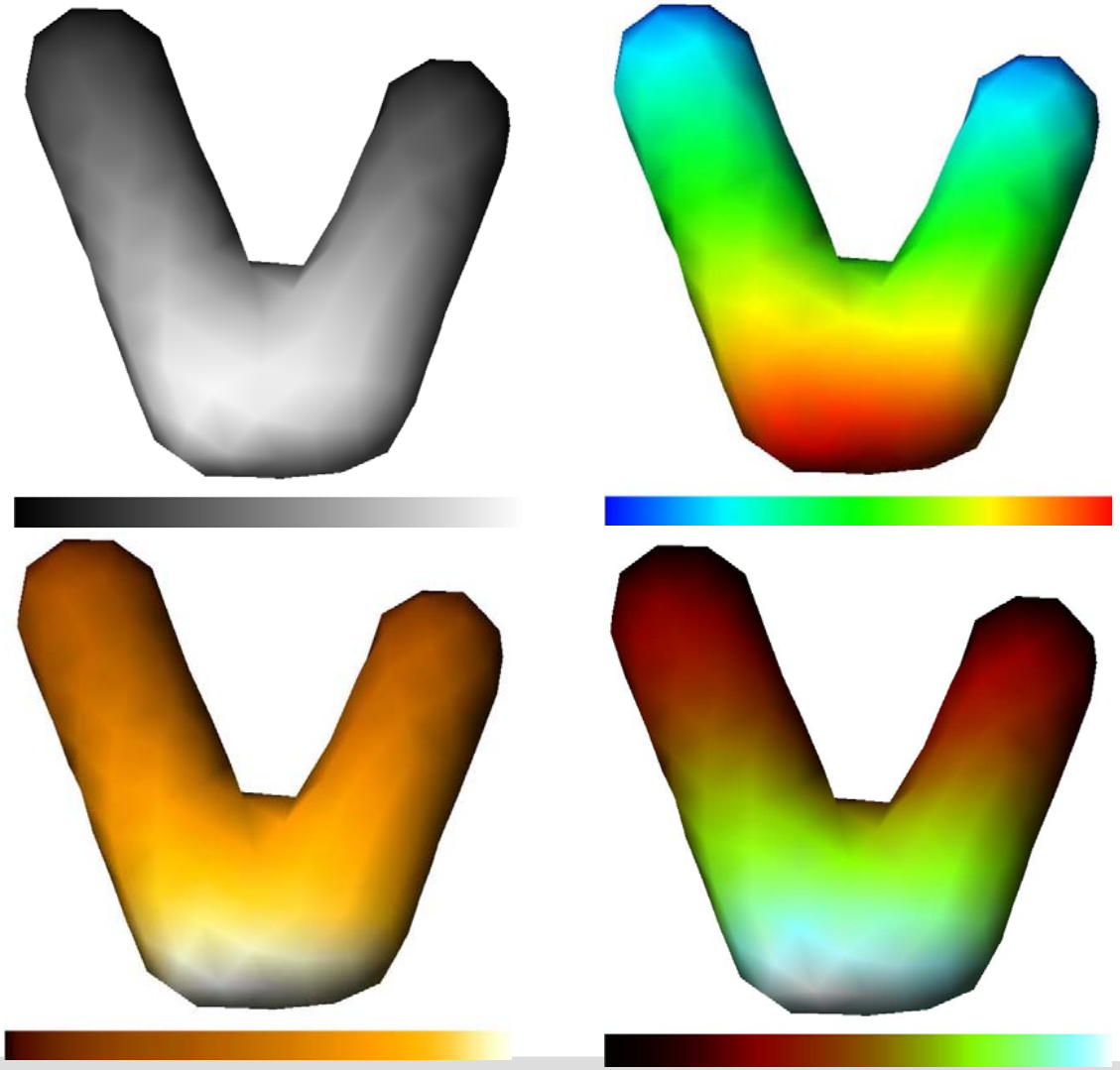
Color order shall be based on human perception.

Rainbow vs. luminance map

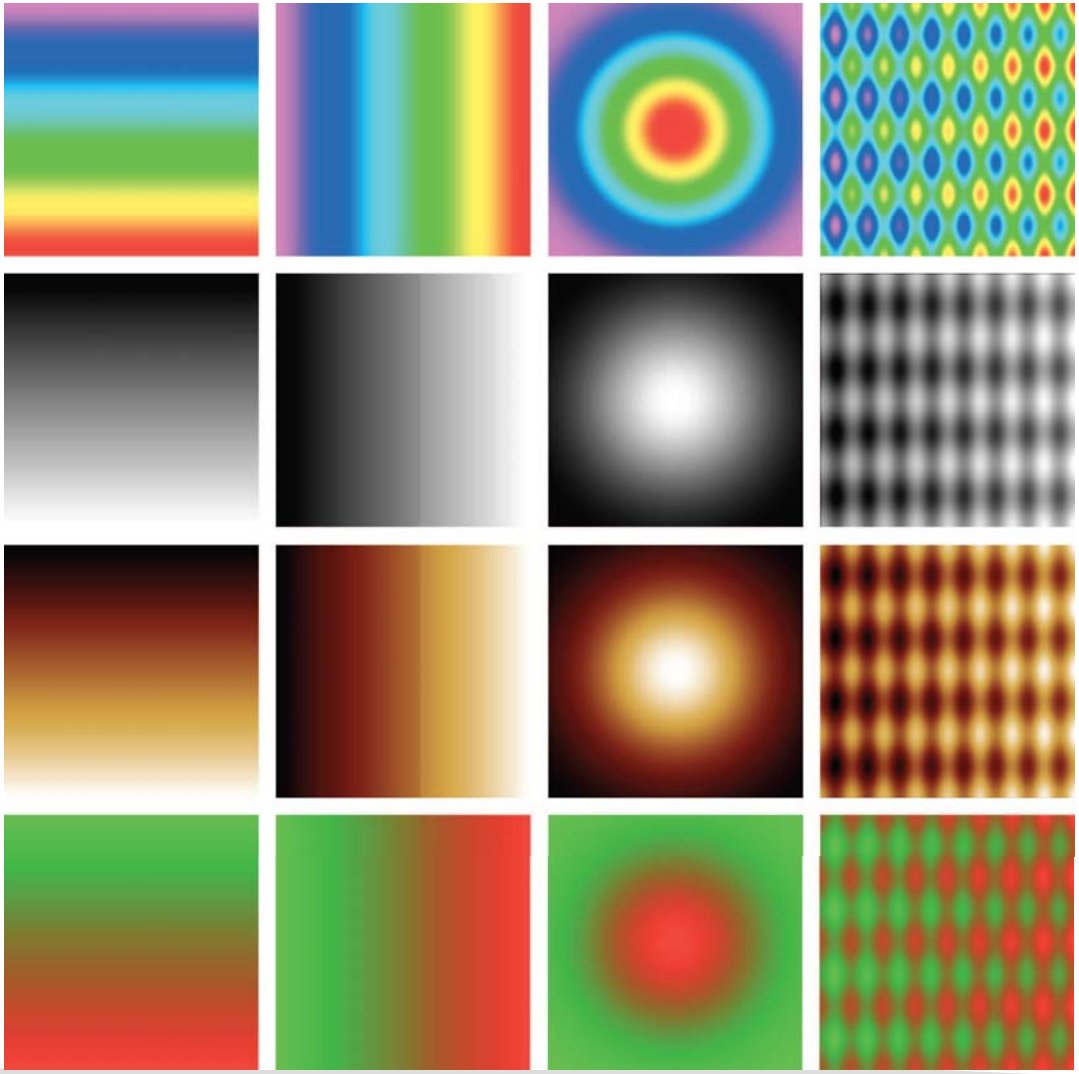


Luminance provides better high-frequency separation than hue.

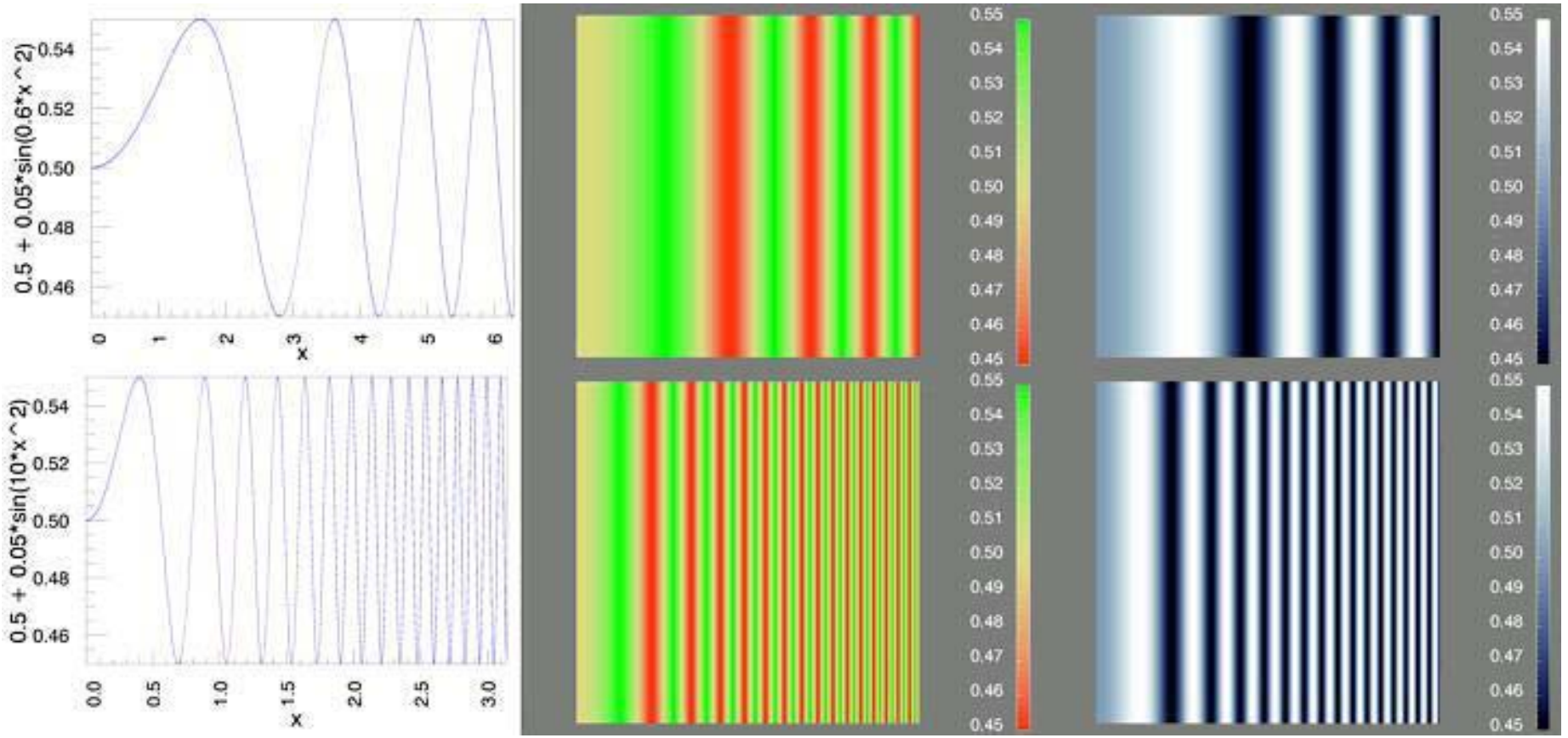
Using luminance and hue



Comparing maps with changing luminance and hue



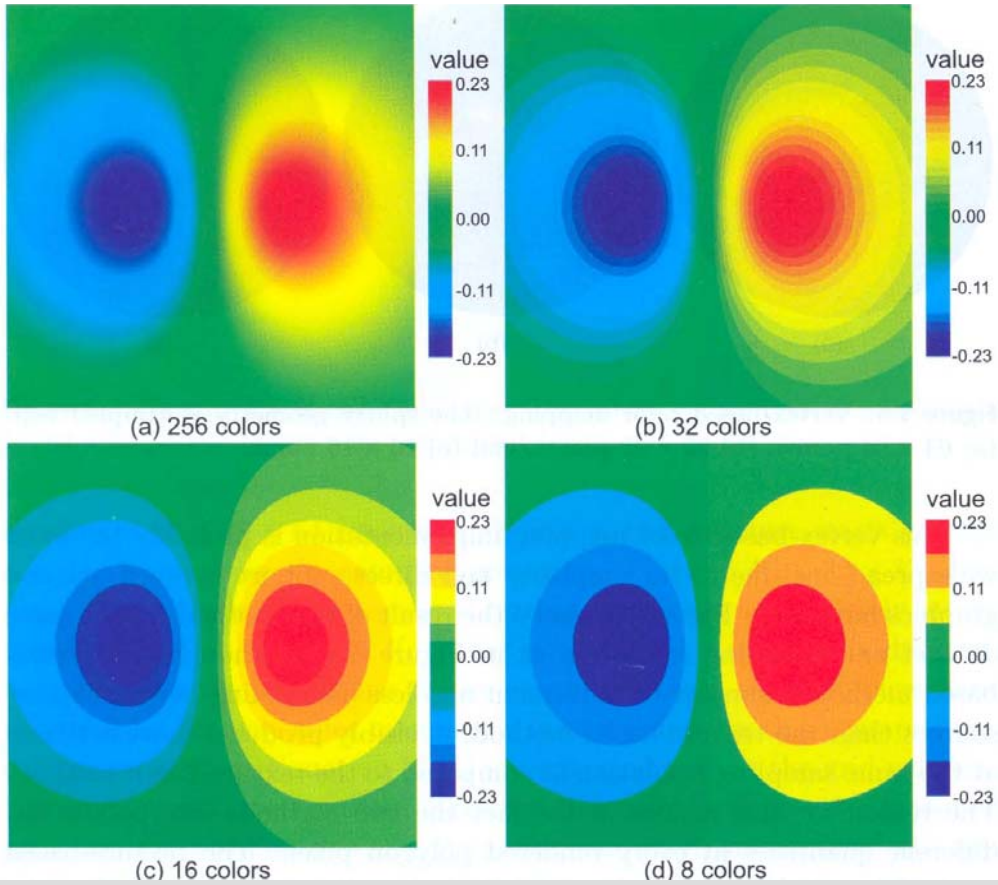
Saturation vs. luminance variation



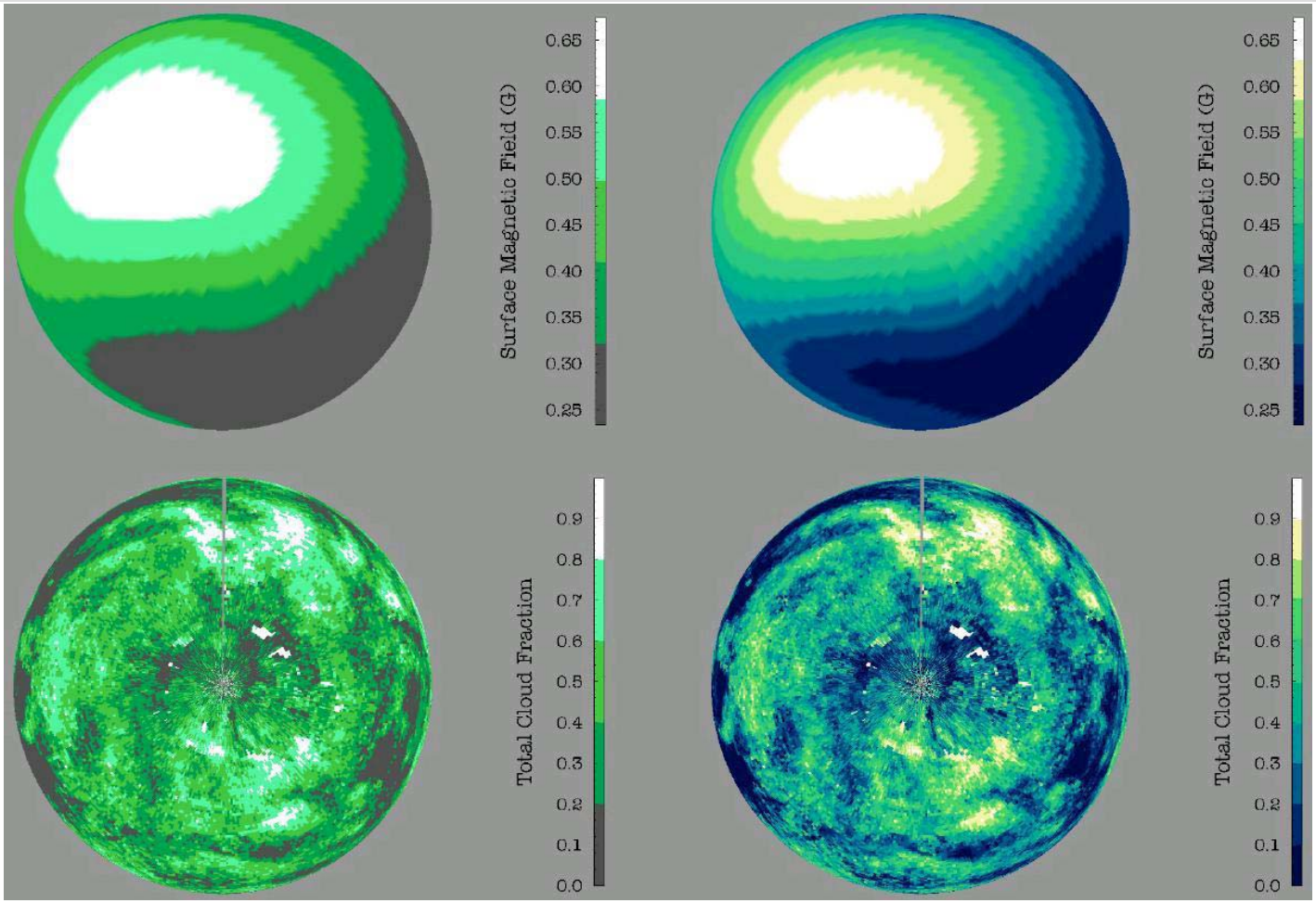
- Saturation change exhibits low-frequency changes better
- Luminance change exhibits high-frequency changes better

Color bands

- The transitions do not need to use the full color spectrum.
- Color reduction leads to a banding effect.

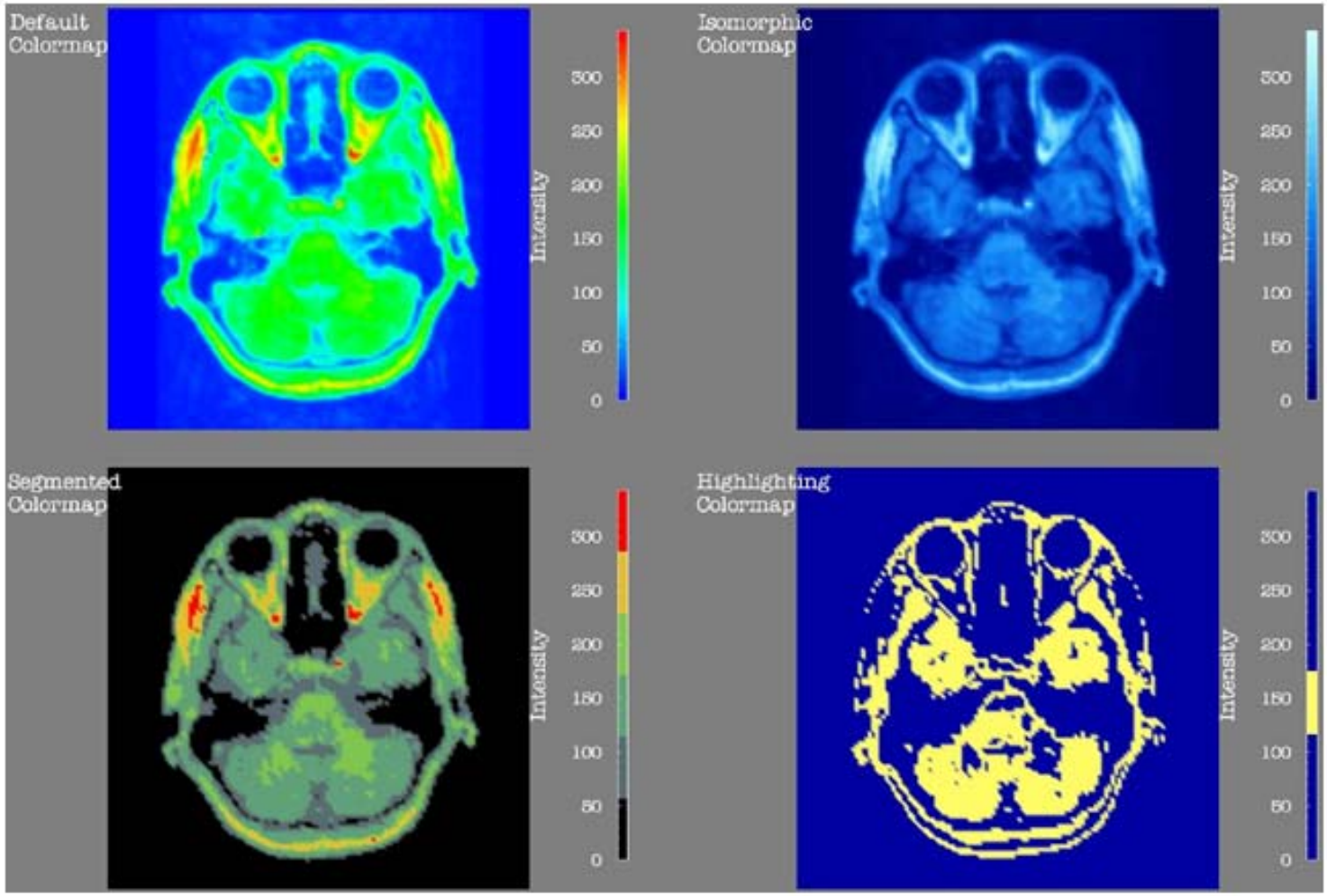


Number of bands



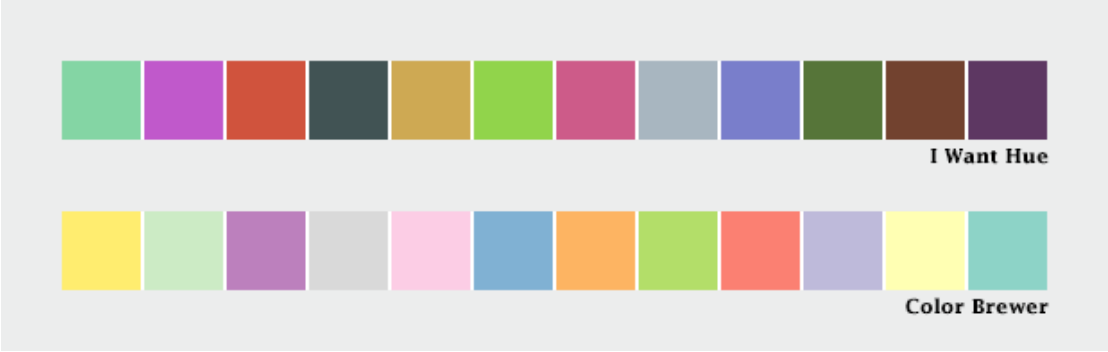
Only for low-frequency changes, additional bands can help

Highlighting color map



Summary

- Numerical data:
  - continuous color transition
  - consider perception (luminance, saturation)
- Ordered data:
  - banded color transition
- Categorical data:
  - distinct colors



[Tool: Color Brewer]

Outro

What we have learned …

- Definitions of Big Data and Data Analytics
- Taxonomy
- Visual Analytics
  - Visual Encoding
  - Interaction Mechanisms
  - Tools
  - Color Models & Color Mapping
- Multidimensional Data
  - Dimensionality Reduction
    - Linear Approaches
    - Non-linear Approaches
  - Interactive Analysis
- Clustering
- Supervised Learning



What we have learned …

- Text and Document Analytics
- Image Analytics
- Relations
  - Graphs
  - Hierarchies
- Temporal Data Analytics
- Geospatial Data Analytics
- Geospatio-temporal Data Analytics



The End





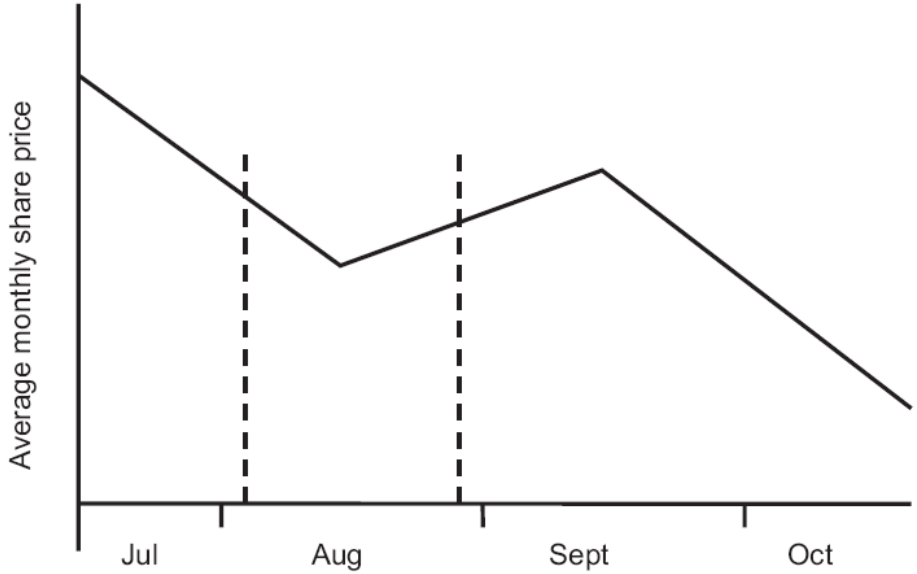
Bonus Material:
Exemplary questions for final preparation

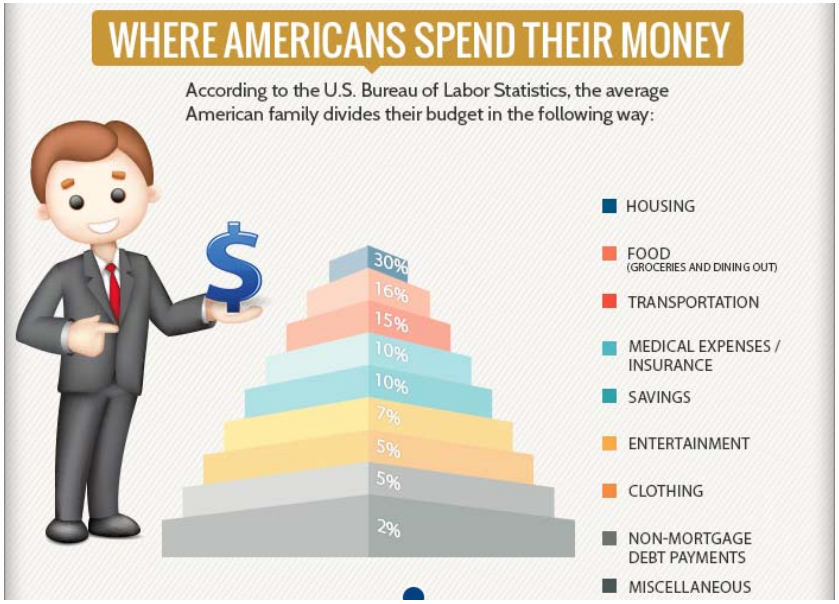




Example 1

What is conceptionally wrong with these visual encodings?





Example 2

Consider a data set on computer monitor models.

Each model is characterized by the following attributes:

company, release date, price, resolution, and quality-test rating.

- Characterize the attributes by declaring of which type each attribute is.
- Design a suitable visual encoding for each of the following subset of attributes:
  - quality-test rating,
  - release date and price,
  - company, price, resolution, and quality-test rating,
  - release date, price, and resolution.

  Explain your design choices.



Example 3

Given a three-dimensional data point with numerical attribute values (0.5,0.2,0.3).

Show how the data point is mapped to:

- Star coordinates and
- Parallel coordinates.

Make sure you label and scale your axes.



Example 4

Explain the duality of Cartesian coordinates and parallel coordinates for two dimensions.

In particular, explain how collinear points in Cartesian coordinates are mapped to parallel coordinates and why.



Example 5

Explain the interaction mechanisms of Reconfigure, Encode, and Filter.
Provide examples for each of them.



Examples (Prof. Wilhelm)

1. Briefly explain the difference between supervised and unsupervised learning!
2. Briefly explain cross-validation.
3. Briefly explain the main idea of bagging!
4. What is the purpose of pruning a tree?
5. Briefly explain how a regression tree works!



The Very End



